Ref / Title	Dataset Description	Methods	Performance	Research Gap	Pros	Cons
[1] putEMG—A Surface Electromyography Hand Gesture Recognition Dataset (Kaczmarek, Mańkowski, Tomczyński, *Sensors*, 19(16):3548, 2019) DOI 10.3390/s19163548	putEMG (https://biolab.put.poznan.pl/putemg-dataset/); 44 able-bodied subjects, 2 sessions $\geq$1 week apart; 24-channel sEMG (3 bands $\times$ 8 electrodes) at 5120 Hz, 12-bit; 8 classes (idle, fist, flexion, extension, 4 pinches); 20 repetitions per gesture per session; ground truth from video	Band-pass 20–700 Hz + adaptive notch; 488 ms windows, 244 ms step; feature sets RMS, Hudgins (MAV, WL, ZC, SSC), Du (IAV, VAR, WL, ZC, SSC, WAMP); LDA, QDA, kNN ($k = 5$), SVM-RBF ($C = 50$); 3-fold validation within each subject–day (88 subsets)	Best mean accuracy $\approx$90% (SVM + RMS); LDA + Hudgins 89%, LDA + Du 88%, kNN + RMS 87%, QDA + RMS 81%; top three not significantly different (ANOVA + Tukey, $p < 0.05$)	No large, open, multi-session sEMG dataset with accurate ground truth existed for benchmarking gesture recognition; the paper provides one and sets baseline results for classic feature–classifier pairs	Large subject count and two sessions allow cross-session studies; video-based labels correct for reaction delay; data and code are public; 3-band layout supports channel-subset studies	Only shallow classifiers; validation is within subject and day, so no cross-subject or cross-session results; $\approx$0.5 s windows are long for real-time control; able-bodied subjects only
[2] EMG Pattern Recognition via Bayesian Inference with Scale Mixture-Based Stochastic Generative Models (Furui, Igaue, Tsuji, *Expert Systems with Applications*, Art. 115644, 2021)	putEMG, session 1 of all 44 subjects, used as three datasets (repeats_long, repeats_short, sequential)	Middle electrode band only (8 channels); 7 active gestures; each putEMG task evaluated as a separate dataset. Rectification + 2 Hz 2nd-order Butterworth low-pass; finite mixture of multivariate scale-mixture models trained by variational Bayes ($K = 10$ with pruning, ν set by mutual information); baselines GMM, LDA, GNB, ν-SVM, MLP, LLR, kNN; train on 2 trials, test on the rest	Proposed: 77.45% (repeats_long), 78.71% (repeats_short), 75.15% (sequential); best baseline ν-SVM 76.11 / 76.85 / 73.37%; LDA 71.77 / 72.83 / 69.49% (paired t-test, Holm correction)	Conventional classifiers ignore the variability of EMG variance, which limits accuracy; the paper models it with a scale-mixture generative model	Probabilistic model with built-in uncertainty; gains are statistically tested on all three putEMG tasks; works with only 8 channels and 5% of training samples	Offline evaluation only; first session only, so no cross-session test; 16 of 24 channels unused; transition periods excluded; accuracy depends on ν, whose best value varies per subject
[3] A Multi-Scale CNN for Transfer Learning in sEMG-Based Hand Gesture Recognition for Prosthetic Devices (Fratti, Marini, Atzori, Müller, Tiengo, Bassetto, *Sensors*, 24(22):7147, 2024) DOI 10.3390/s24227147	NinaPro DB2, DB3 (amputees) and DB7; 14 activities-of-daily-living gestures selected from NinaPro's 40; 15 subjects (12 able-bodied, 3 amputees) held out as new users for transfer learning	4th-order Butterworth band-pass 10–500 Hz + 50 Hz notch; windows 100–250 ms with 75% overlap (main: 150 ms); multi-scale CNN (parallel and separable convolutions, batch norm, dropout, 1×1 fusion); fine-tunes last ($\approx$70k params) or middle ($\approx$200k) layers	97% F1 over 14 classes in 1.40 epochs on average; 98% $\pm$ 0.02 accuracy on a new subject (middle fine-tuning, 2 epochs, 4 repetitions, 150 ms window); $\approx$3 ms inference on a Tesla V100	Models trained on one set of users do not work well for a new user, and collecting many repetitions from each user is impractical; the paper uses transfer learning to adapt to a new user, including amputees, with few repetitions	Short windows suit real-time prosthesis control; includes amputee data; adapts to a new user in about 2 epochs; inter- and intra-subject splits both reported	Not evaluated on putEMG; weakest on amputee data (DB3); timing measured on a server GPU, not an embedded device; very high DB7 accuracy may reflect overfitting
[4] Big Data in Myoelectric Control: Large Multi-User Models Enable Robust Zero-Shot EMG-Based Discrete Gesture Recognition (Eddy, Campbell, Bateman, Scheme, *Frontiers in Bioengineering and Biotechnology*, 12:1463377, 2024) DOI 10.3389/fbioe.2024.1463377	EMG-EPN612 (public): 612 users, Myo armband (8 ch, 200 Hz), 6 gestures (rest, wave in, wave out, hand open, hand close, double tap); 306 training users with 50 reps, 306 test users with 25 labelled reps. Plus new Myo DisCo set: 14 users, 2 sessions 1 to 7 days apart, 4 extra limb positions, slow/fast speeds, 4 out-of-set gestures	Discrete control (one label per whole gesture). MVLDA (200 ms window, 25 ms step), 1-NN DTW and a 3-layer LSTM (128 units) on 25 ms RMS windows; features RMS, MAV, WENG, MNP, WL, HTD, LS4. Zero-shot cross-user training; traditional and contrastive transfer learning (last layers only); softmax-confidence rejection; ANOVA / Friedman tests	LSTM zero-shot: 93.0% on 306 unseen users vs $\sim$97.9% user-dependent; LSTM beats MVLDA by 17 to 24% and DTW by $\sim$18%. 90% needs $\sim$150 users $\times$ 30 reps. Transfer learning with 10 reps: $\sim$97% vs $\sim$95% user-dependent. New dataset: $\sim$93% on day 1, $\sim$3% higher on day 2; slow gestures lose 8 to 11%. Rejection at 99.99%: 92.6% to 94.5% (4.4% rejected)	Zero-shot cross-user myoelectric control was assumed infeasible; earlier cross-user work used small cohorts (under 50 users) and left open how many users, which features and which biases matter	Largest public cohort; ten mini-studies (user count, window, features, gesture set, bias, transfer); tests a second dataset with cross-day, limb-position and speed confounds; open data and LibEMG toolkit	Myo only (discontinued, 200 Hz); offline, no user-in-the-loop test; 6 discrete gestures; training data skewed (66% men, 96% right-handed) so women, left-handed and slow users score lower; out-of-set rejection weak (23% rejected); unimpaired users only
[5] EMG Dataset for Gesture Recognition with Arm Translation (Kyranou, Szymaniak, Nazarpour, *Scientific Data*, 12:100, 2025) DOI 10.1038/s41597-024-04296-8	New dataset: 8 able-bodied men; 16-ch sEMG (4 Delsys Trigno Quattro, 2 kHz) plus 18-sensor CyberGlove II kinematics; 6 grasps (power, lateral, pointer, tripod, open, rest); 9 forearm positions on a 3×3 grid (45° steps from neutral); 5 reps per grasp and position (5 s hold, 3 s rest); 2 consecutive days, 2 sessions per day	20 to 450 Hz 4th-order Butterworth + notch; 128 ms windows, 50 ms stride; Hudgins features (MAV, ZC, SSC, WL); LDA; 80:20 trial split with 5-fold CV. Four baselines: within-position, naive cross-position transfer, position classification per grasp, hierarchical multi-label classifier (HMC)	$\sim$96% within one position; up to 10% drop when train and test positions differ (neutral position best); position itself classified at $\sim$80% per grasp; HMC position encoder 58%, grasp accuracy 92%+ when position correctness is ignored	Few public datasets capture arm position; the three existing ones cover mostly a single vertical plane, so position-invariant myoelectric control has no standard benchmark	First 3×3 position grid (vertical + horizontal planes); balanced across positions; EMG with glove kinematics; two days of variability; data (Dryad), collection code and example pipeline public	Only 8 male able-bodied subjects; 6 grasps and 5 reps; only LDA + Hudgins baselines; no amputees or women; static holds are fatiguing and recording length is limited by sensor hardware; HMC baseline performs poorly
[6] Gesture Detection from sEMG Signals Based on Similarity Learning (Celik, Bayazit, *Neural Computing and Applications*, 37(22):18663–18677, 2025) DOI 10.1007/s00521-025-11393-1	Myo dataset of Côté-Allard et al.: 8-ch Myo (200 Hz), 7 gestures; 18 subjects for pretraining, 17 subjects for fine-tuning and testing (1 to 4 cycles of fine-tune data). Also NinaPro DB5 Exercise B (10 subjects, one Myo, 18 classes), leave-one-subject-out	Pretrained 2D CNN (2 conv layers of 48 filters + FC) on 52-sample (260 ms) windows with circular sensor padding; multi-pass fine-tuning with progressive layer freezing; CNN features fed to an ϵ-SVR that learns pairwise similarity; kNN ($k = 15$) voting. Ablation, one-tailed t-tests	95.87% (1 cycle) to 98.77% (4 cycles) on 7 classes, 17 test subjects; beats CWT+transfer learning (94.69% at 1 cycle, 22% fewer errors); significant at 1 to 3 cycles, not at 4. DB5 Ex. B: 82.05% (18 classes). 20.14 ms per window on one CPU core; INT8 TensorRT loses under 0.01%	Personalised gesture recognition needs high accuracy from very little user data; most transfer learning only fine-tunes a classifier and ignores sample similarity	Works with a single calibration cycle; latency well inside the 40 ms budget; each component ablated; two datasets; quantisation tested	Offline only; both datasets are 8-ch Myo with small cohorts; gain not significant with 4 cycles; modest DB5 accuracy; per-user SVM and kNN training; able-bodied subjects only; code on request
[7] Posture-Invariant Myoelectric Control with Self-Calibrating Random Forests (Jiang, Ma, Nazarpour, *Frontiers in Neurorobotics*, 18:1462023, 2024) DOI 10.3389/fnbot.2024.1462023	Pre-training: 66 participants from three own datasets (8 Delsys Trigno electrodes, 6 gestures, arm pointing down). Real-time test: 20 new participants; 1 calibration trial (2 s) per gesture at elbow 90°, then 11 blocks over 5 arm positions (rotating clockwise and counter-clockwise), 5 trials per gesture per block	Random forest: 200 trees pre-trained on z-scored features (200 ms window, 100 ms step; MAV, WL, RMS, SSC, ZC, skewness, PKF, MDF, MNF, VCF; 80-D vector). One-shot fine-tuning by pruning trees and appending 200 user-specific trees. Unsupervised self-calibration after each block (t-SNE + K-Means pseudo-labels, 1,500-sample buffer, 40% of appended trees replaced). Baselines LDA, SVM, standard RF; Friedman + Nemenyi tests	Trained and tested at the same position: LDA 76.0%, SVM 76.7%, RF 77.6%. At other positions LDA drops up to 6.4%, SVM up to 5.1%, RF under 1.5%. Pre-training + fine-tuning adds 2.5 to 4.2%; self-calibration adds a further 3.3 to 5.0%. Full model 84.6% $\pm$ 8.5 (ablation replay); pseudo-label accuracy 80.1%; removing energy features drops it to 71.4%	Arm position shifts EMG and can cut accuracy by up to $\sim$40%; fixes need multi-position training data, extra sensors or large deep networks; a light, explainable, position-invariant model that needs little user data is missing	Real-time test (100 ms decisions) with 20 users during arm rotation; one 2 s calibration trial per gesture; simple, explainable, parallelisable model; ablation of every component; hyper-parameters tuned only on pre-training users	Limb-intact users only; 6 gestures and 5 arm positions; open-loop (no feedback to users); baselines run offline; no deep-learning comparison; absolute accuracy only $\sim$85%; data on request only
[8] Replay-Based Incremental Learning Framework for Gesture Recognition Overcoming the Time-Varying Characteristics of sEMG Signals (Zhang, Li, Sun, Zhang, Zhang, Zhang, *Sensors*, 24(22):7198, 2024) DOI 10.3390/s24227198	Own dataset: 8 healthy adult men (20 to 28 y); 4-ch forearm sEMG (EMG06 module, 1000 Hz); 7 wrist/hand gestures (extension, flexion, radial deviation, ulnar deviation, supination, pronation, fist); 12 consecutive days, 100 reps per gesture per day; public on FigShare	20 to 300 Hz band-pass + 50 Hz notch; energy-threshold segmentation to 1000-sample segments; 1D DenseNet (4 dense blocks, growth rate 24); replay-based incremental learning with weighted loss ($\lambda \geq 0.2$); replay samples chosen by DBSCAN on RMS vs random, uniform, intermediate; class-incremental learning; per-subject training	Single-day >99% (DenseNet 99.8%) but cross-day only 64.2% (70.6% at 1-day gap, 45.5% at 11 days). After 9 increments 96.5%; DBSCAN replay 93.7% with 140 samples; incremental 95.5% vs 94.6% full-data training, $\sim$10 s vs 64 s. Class-incremental 88.6% (95.2% before adding a class)	sEMG accuracy decays across days from electrode displacement and signal drift; storing all old data is costly and retraining from scratch is slow	Quantifies cross-day decay; replay keeps training time near 10 s; DBSCAN selection works with few samples; can add new gestures; four CNN backbones compared; data public	Only 8 male subjects, 4 channels, 7 gestures; 12-day span is short; per-subject models; adding classes lowers accuracy (95.2% to 88.6%); fixed loss weight; offline evaluation
[9] Context-Informed Incremental Learning Improves Both the Performance and Resilience of Myoelectric Control (Campbell, Eddy, Bateman, Côté-Allard, Scheme, *Journal of NeuroEngineering and Rehabilitation*, 21:70, 2024) DOI 10.1186/s12984-024-01355-4	No public dataset; two online experiments with the Myo armband (8 ch, 200 Hz), 32 participants: Exp. 1, 11 users with only 1 s of training per class; Exp. 2, 21 users after a 45° electrode shift. 5 classes (wrist flexion, extension, hand open, close, neutral) mapped to cursor moves in a target game, then a Fitts' law test	LDA on Hudgins time-domain features (200 ms window, 100 ms step), adapted online in 10 s batches with Context-Informed Incremental Learning: negative (N), positive (P) and P+N pseudo-labels from task context, vs unsupervised high-confidence (UHC) and screen-guided training (SGT). 2 min (Exp. 1) and 5 min (Exp. 2) adaptation; Friedman + Finner tests; NASA TLX	Exp. 1 accuracy: SGT 91.28%, P 86.01%, P+N 85.26%, UHC 79.05%, initial 65.75%; throughput P 0.43 bit/s vs initial 0.24. Exp. 2 after shift: no adaptation 48.32% vs pre-shift 87.78%; P+N 79.53% (not significantly different); P+N throughput 0.61 vs pre-shift 0.50 bit/s; only 5 of 21 users recovered with UHC, all with N-CIIL	EMG control degrades with use; unsupervised adaptation trusts its own predictions and can reinforce errors, and offline calibration misses real user behaviour	Online user-in-the-loop test with 32 people; works from minimal data and after electrode shift; beats screen-guided training on throughput; subjective workload measured; code and data public (GitHub)	Simple 2-DOF game with perfect task context; 5 classes and LDA only; needs context to be available; limb-intact users; offline accuracy and throughput disagree; N-CIIL may plateau; not comparable to benchmark dataset accuracies